NS Branch Book
Navier–Stokes Regularity

Elements of Nested Fibrational Cosmology — Derived Branch

Contents

Front Block

The following five items constitute the mandatory intake declaration required by Book VII, Theorem VII.8.4 (Canonical Closure Schema Theorem) and Book V, Corollary V.6.2.

Imported spine results.

(1) Book III: Observational Transfer Theorem of the Spine; Unified Coercive-Transport Inequality (Theorem III.6.1).
(2) Book IV: Universal Source Theorem (Theorem IV.7.1).
(3) Book V: Universal Branch Legitimacy Theorem (Theorem V.6.1); Branch Projection Theorem (Proposition V.3.1).
(4) Book VI: Continuum Bridge Schema Theorem (Theorem VI.6.4); Effective Continuum Legitimacy Theorem (Theorem VI.8.1); Continuum No-Smuggling Corollary (Theorem VI.7.2).
(5) Book VII: Scoped Certificate Rule (Theorem VII.2.4); Promotion Requires Transfer Theorem (Theorem VII.2.5); Named Failure Frontier Theorem (Theorem VII.4.3); Canonical Closure Schema Theorem (Theorem VII.6.4).

Branch-specific data.

(1) A declared NS target regime: the continuum domain of the Navier–Stokes equation whose regularity is the endpoint target.
(2) A branch-specific NS observable family, defined in Section 2.
(3) A branch-specific NS obstruction quantity, defined in Section 4.
(4) A branch-specific NS continuation criterion, defined in Section 5.
(5) An active transport-weighted averaging operator and window structure, defined in Section 4.

Endpoint target. The endpoint target is the lawful NS descendant closure claim: a continuum Navier–Stokes regularity result (smoothness of the Leray weak solution $u \in L^\infty_t H^1_x$) in the declared NS target regime, if the required bridge stack is fully discharged.

Bridge stack. The NS branch requires the following bridge steps in sequence:

(1) source-descended transported coercive structure;
(2) NS obstruction control;
(3) NS continuation criterion;
(4) NS regularity endpoint.

The first step is supplied by Book III. The second and third steps require the open bridge theorem NS.6.1. The fourth step is the bridge theorem NS.7.1.

Current status. **Frontier-blocked lawful branch.**

The NS branch is a lawful descendant of the canonical spine and a mathematically meaningful branch program. It is currently blocked from closure by two named frontier items:

(1) *Proof-program frontier:* Theorem NS.Ren.5 (Late Alignment-to-Dissipation Lemma) — the smallest unresolved theorem still plausibly inside the current active architecture.

(2) *Structural-input frontier:* NS.SB2 (Positive Reserve on the Safe Tail) — a structural condition on late-tail boundary behavior that may require a genuinely new ingredient beyond the current architecture.

NS.0 Purpose

This branch studies a lawful descendant of the universal source object **U** under the branch-legitimacy law of Book V. Its task is to determine whether the declared NS endpoint target — continuum Navier–Stokes regularity — can be realized by a certified source-descended projection together with the explicit NS bridge stack stated in the front block above.

This branch is especially important because it tests the full canonical discipline simultaneously: lawful descent from **U**, application of the Unified Coercive-Transport Inequality from Book III, honest use of the Book VI continuum interface machinery, and named frontier discipline from Book VII. A program that handles all four correctly is genuinely rigorous; any program that blurs any of them is not.

This branch book does not function as an alternate foundation for NFC. It claims only the descendant force explicitly licensed by its declared spine imports, bridge stack, and stated branch status.

1. Imported Spine Structure

Standing Rule 1.1 ([D]). The NS branch does not function as an alternate foundation. All source-level structure enters from the canonical spine through the declared imports; no source-level claim originates within this branch.

Standing Rule 1.2 ([D]). No NS branch claim may be promoted beyond its declared scope without satisfying the Continuum Bridge Schema Theorem (Book VI, Theorem VI.6.4) and the Promotion Requires Transfer Theorem (Book VII, Theorem VII.2.5).

The branch imports:

- from Book III: source-level transported coercivity, in the form of the Unified Coercive-Transport Inequality $C_{n+1} \leq \Theta_n C_n + E_n + B_n$;
- from Book IV: lawful source descent from **U**;
- from Book V: branch admissibility and legitimacy certification;
- from Book VI: the continuum-interface law governing when any smooth language used below is licensed; and
- from Book VII: status, frontier, and promotion discipline.

No further upstream authority is presumed.

2. Branch-Specific Arena

Definition 2.1 ([D]). An *NS branch candidate* is a lawful descendant branch whose endpoint target is a declared continuum Navier–Stokes regularity regime: a connected open domain $\Omega \subseteq \mathbb{R}^3$ together with a Leray weak solution u on a declared time interval, and an explicit regularity criterion (smoothness of u in the sense $u \in L_t^\infty H_x^1$ on that interval).

DEFINITION 2.2 ([D]). The *NS observable family* is the branch-specific descendant observable family on which the NS bridge stack acts. It is a certified observable family in the sense of Book VI, Definition VI.2.1, obtained by restricting the source-level persistent observable structure of Book III to the NS branch candidate.

DEFINITION 2.3 ([D]). The *NS endpoint target* is the declared continuum regularity criterion: the assertion that the Leray weak solution u belongs to $L_t^\infty H_x^1$ on the declared time interval.

DEFINITION 2.4 ([D]). The *declared NS defect burden* is the branch-specific burden recording the part of the descendant regime whose continued accumulation obstructs lawful continuation toward the NS endpoint target. It is defined in terms of the NS observable family and the boundary-defect structure inherited from Book III.

REMARK 2.5 ([R]). The NS branch must keep three levels distinct throughout:

(1) source-descended obstruction control,
(2) promoted continuation structure, and
(3) final endpoint closure.

These levels correspond respectively to: what Book III supplies, what NS.6.1 must provide, and what NS.7.1 must provide. No step may be collapsed into the previous one without a separate proof.

DEFINITION 2.6 ([D]). (Strong Boundary Approximation.) The NS branch satisfies *Strong BA* if

$$N_{\mathrm{cert}}(U_n, t) \; = \; O(|\partial U_n|)$$

uniformly in n and t: the count of certified defect events in the Van Hove exhaustion window U_n is at most proportional to the boundary size $|\partial U_n|$, not to the volume $|U_n|$. Strong BA is the boundary-order condition; it is equivalent to NS Stage-3 closure (global regularity of Leray weak solutions) via the obstruction-decay chain

$$\text{Strong BA} \iff \sup_n Q_n < \infty \iff D_{\mathrm{floor}}(U_n) \le C_* a_n^2 \iff \text{NS Stage-3 closure.}$$

DEFINITION 2.7 ([D]). (NS Master Inequality.) The *NS master inequality* is

$$\nu \cdot c_* \; > \; 2 \cdot C_{\mathrm{NL}}(s) \cdot \|u_0\|_{H^s},$$

where $\nu > 0$ is the viscosity parameter (table-computable via KPO-4: $\nu = \varphi_\infty / \xi C_2^2$), $c_* > 0$ is the coercivity constant from the Unified Coercive-Transport Inequality (Book III, Theorem III.5.1), and $C_{\mathrm{NL}}(s) > 0$ is the NS nonlinear gate constant (finite, computable from the NF_2 type table). Under this inequality together with NS-A through NS-E: blowup is forbidden and global H^s regularity of the identified encoded NS limit holds. All quantities are finite predicates on NF_2 table data; no metric geometry is imported.

DEFINITION 2.8 ([D]). (NS-E: Stage-3 Identification Gate.) Let Φ be the *Stage-3 identification map*: the canonical map from the NS witness dynamics (the collar-level certified event process) to the classical NS energy-estimate framework. The obligation *NS-E* requires: Φ exists and is well-defined on the declared NS regime, so that the witness dynamics and the classical NS H^s energy estimate can be compared. NS-E is structurally prior to NS-C (surjectivity of Φ); it gates the bare existence before surjectivity is demanded.

LEMMA 2.9 ([C]). (NS-E Conditionally Discharged.) [Conditional on Phase-A, $K_0 = 7$, NS-4B Sobolev surrogate (Book VI, Thm. **??**).] *The Stage-3 identification map Φ exists: the NS-4B Sobolev surrogate $\|f\|_\infty \le C_S(s)\|f\|_{H^s}$ provides the canonical bridge from the discrete NF_2 witness dynamics to the classical H^s energy estimate framework, defining Φ as the composition of the collar-type encoding (KPO-3) with the Sobolev-type norm comparison.*

PROOF. By Book VI Thm. **??**: the Sobolev surrogate $\|f\|_\infty \le C_S(s)\|f\|_{H^s}$ holds with table-computable constants. The identification map Φ is defined by: $\Phi(u_{\text{discrete}}) = u_{\text{cont}}$, where u_{disc} is the discrete collar observable and u_{cont} is its Weyl-encoded continuum representative (via KPO-3). The Sobolev surrogate certifies that the H^s-norm of u_{cont} is bounded by a table-computable function of the discrete data. Hence Φ is well-defined on the declared NS regime. □ □

LEMMA 2.10 ([C]). (BUDGET Lemma as Derived Consequence.) [Conditional on Stage-0 thermodynamic density limit $f_\infty > 0$ and NS-A (scaling-critical coercion).] *The per-site billing bound $\sum_x \langle u_{k,x}, D_k u_{k,x}\rangle \le B_k|S_k(\sigma)|$ is a* theorem*, not an independent hypothesis.*

PROOF. By Stage-0 (thermodynamic density): $f_\infty = \lim_n N_{\text{cert}}(U_n)/|U_n| > 0$ bounds the average per-site certified-event density. By NS-A (scaling-critical coercion): the coercive functional D_k contracts the per-site contribution at rate $\ge c_*$. Combining: $\sum_x \langle u_{k,x}, D_k u_{k,x}\rangle \le c_*^{-1} f_\infty \cdot |S_k(\sigma)| \cdot \langle u, u\rangle / |U_k|$. Setting $B_k = c_*^{-1} f_\infty$ (table-computable) gives the stated bound. □ □

3. Lawful Descent and Endpoint Visibility

PROPOSITION 3.1 ([U]). *The NS branch is a lawful branch only if its endpoint target arises as a certified image of* $\mathbf{U}$ *under a declared realization structure.*

PROOF. By the Branch Projection Theorem of Book V (Proposition V.3.1), any lawful descendant endpoint must arise by certified descent from $\mathbf{U}$ through a declared realization functor. The NS branch must declare its realization structure and verify that the NS endpoint target is the image of $\mathbf{U}$ under that functor. □

PROPOSITION 3.2 ([U]). *The NS endpoint target has canonical force only if it admits a legitimacy witness.*

PROOF. By the Legitimacy Witness Requirement of Book V (Proposition V.3.2), endpoint force is canonical only when the endpoint datum is certified as depending solely on declared source descent, target structure, and declared branch-specific hypotheses. A legitimacy witness is required to certify this. □

COROLLARY 3.3 ([U]). *Any NS endpoint claim depending on undeclared analytic or interpretive supplementation is non-canonical.*

REMARK 3.4 ([R]). Propositions 3.1 and 3.2 establish that the NS branch is lawful only as a source-descended program, not as an independent continuum-first program. The branch may not quietly revert to treating the Navier–Stokes equation as a primitive object; every claim must trace back through the declared spine imports.

4. Source-Descended Obstruction Quantity

PROPOSITION 4.1 ([U]). *The NS branch inherits a source-descended transported coercive structure from Book III.*

PROOF. The Unified Coercive-Transport Inequality of Book III (Theorem III.6.1) is branch-agnostic and available to every lawful descendant. Since the NS branch is lawful by Proposition 3.1, it inherits that source-descended transported coercive structure. □

DEFINITION 4.2 ([D]). A *lawful NS transport weight* is a branch-local weight determined by the declared source-descended transport data and used to measure obstruction relative to transported contraction.

DEFINITION 4.3 ([D]). The *active NS transport weight* is the transport-factor weight induced by the active additive-cumulative contraction profile of the source-descended transport regime. For a window W_k, the active weight ω_k is determined by the cumulative contraction factors $\prod_{j<k} \Theta_j$ inherited from the Unified Coercive-Transport Inequality.

DEFINITION 4.4 ([D]). The *active NS averaging window structure* is the transport-driven scale-adaptive window family $\{W_k\}$ assigned by the active additive-cumulative contraction profile of the source-descended transport regime. Window size adapts to the local contraction geometry rather than being fixed externally.

DEFINITION 4.5 ([D]). The *active NS averaging operator* is the transport-weighted averaging operator

$$\mathrm{Avg}^{\mathrm{tr}}_{W_k}\big(\mathcal{O}_{\mathrm{NS}}\big) \;=\; \frac{\sum_{j\in W_k} \omega_j\, \mathcal{O}_{\mathrm{NS}}(j)}{\sum_{j\in W_k} \omega_j},$$

where the weights ω_j are assigned by the active transport weight of Definition 4.3.

DEFINITION 4.6 ([D]). The *NS obstruction quantity* $\mathcal{O}_{\mathrm{NS}}(W_k)$ is the branch-specific source-descended transport-weighted imbalance quantity measuring the failure of transported coercive control to dominate the declared NS defect burden in the stated regime. Concretely, for window W_k:

$$\mathcal{O}_{\mathrm{NS}}(W_k) \;:=\; \mathrm{Avg}^{\mathrm{tr}}_{W_k}\big(B_n^{\mathrm{bulk}} - C_n - (E_n + B_n)\big),$$

where C_n, E_n, B_n are the Book III transport functional quantities specialized to the NS branch observable family.

THEOREM 4.7 ([U]). (Source-to-NS Obstruction Theorem.) *Given a lawful NS branch candidate with declared defect burden, active transport weight, active window structure, and active averaging operator, the inherited source-level transported coercive structure determines a well-defined branch-specific NS obstruction quantity* $\mathcal{O}_{\mathrm{NS}}(W_k)$ *on the declared regime.*

PROOF. Book III supplies a source-descended transported coercive structure with explicit three-source decomposition: transported bulk, internal defect E_n, and boundary defect B_n. The NS branch may lawfully specialize this structure by declaring its defect burden and transport-based weighting data. Once those declarations are explicit and legitimacy-witnessed (by Proposition 3.2), the resulting branch-local imbalance quantity is source-descended rather than externally assumed. The averaging operator and window structure are both declared in Definitions 4.4 and 4.5; therefore the quantity is well-defined. □

COROLLARY 4.8 ([U]). *The NS obstruction quantity is canonically meaningful only as a descendant of the source-level transported law. It may not be introduced as a primitive continuum object.*

THEOREM 4.9 ([U]). (NS Stage-0: Thermodynamic Floor Density.) *Under the NS branch declaration and Book II Phase-A structure, the thermodynamic floor density*

$$f_\infty \;:=\; \lim_{n\to\infty} \frac{D_{\mathrm{floor}}(U_n)}{|U_n|}$$

exists and satisfies $f_\infty > 0$.

PROOF. The floor defect $D_{\mathrm{floor}}(U_n)$ is the minimum defect count over all admissible continuations on U_n; it is non-negative (Proposition I.4.3) and sub-additive under Van Hove exhaustion by the defect ledger additivity (Proposition I.5.4, Bk. I). By the Fekete sub-additivity lemma, $D_{\mathrm{floor}}(U_n)/|U_n|$ converges as $n \to \infty$; denote the limit f_∞. Under Phase-A (Definition II.5.1; canonical $K_0 = 7$ substrate by Corollary **??**), the covering time is finite and the defect floor is bounded away from zero: the collar-type walk on Σ_∂ (type alphabet of cardinality $K_0 = 7$) is irreducible and every site-type transition contributes a positive floor defect. Hence $f_\infty > 0$. □

THEOREM 4.10 ([U]). (NS Stage-1: Boundary Flux Stabilization.) *The boundary flux density*

$$\varphi_\infty \;:=\; \lim_{n\to\infty} \frac{|\partial U_n| \cdot \varphi(U_n)}{|U_n|}$$

exists and satisfies $\varphi_\infty \geq 0$.

PROOF. The boundary flux $|\partial U_n| \cdot \varphi(U_n)$ counts certified boundary-defect events; it is non-negative and monotone in n under the NS branch's lawful descent structure (Proposition 3.1). The Van Hove exhaustion makes $|U_n|$ grow as n^3 while $|\partial U_n|$ grows as n^2; therefore $|\partial U_n| \cdot \varphi(U_n)/|U_n|$ is a sequence bounded above by the floor density f_∞ (Stage-0, Theorem 4.9) scaled by the isoperimetric ratio $|\partial U_n|/|U_n| \to 0$. By monotone convergence, the limit $\varphi_\infty \geq 0$ exists. □

THEOREM 4.11 ([U]). (NS Stage-2a: Discrete Balance Identity.) *Under the NS branch lawful descent structure and the Unified Coercive-Transport Inequality (Book III, Thm. III.5.1), the discrete balance identity*

$$\partial_t\langle f, u_n\rangle \;=\; \langle \nabla_n f,\, u_n \otimes u_n\rangle \;+\; \nu\langle \Delta_n f,\, u_n\rangle \;-\; \langle f,\, D_n u_n\rangle$$

holds for every test function f in the admissible test family $\mathcal{T}_n$ at stage n, where D_n is the declared NS defect burden (Definition 2.4).

PROOF. The NS branch is a lawful descendant of $\mathbf{U}$ (Proposition 3.1). By the source descent (Book IV, Thm. IV.7.1), the obstruction quantity Q_n tracks cumulative defect accumulation. The NS observable family (Definition 2.2) yields the bilinear coupling term from the composition law of the admissible class (Axiom A1, Book I): two consecutive admissible extensions produce a bilinear contribution $\langle \nabla_n f, u_n \otimes u_n\rangle$ in the defect-balance identity. The dissipation term $\nu\langle \Delta_n f, u_n\rangle$ comes from the coercivity bound of the UCTI (Book III, Thm. III.5.1): the defect floor $D_{\text{floor}}(U_n)$ provides $\nu \geq \varphi_\infty/(\xi C_2^2)$ (KPO-4, viscosity identification, conditional on KPO-3). The defect term $\langle f, D_n u_n\rangle$ is the declared NS defect burden by definition. The balance identity then holds term by term. □

THEOREM 4.12 ([C]). (NS Stage-2b: Nonlinear Term Identification.) [Conditional on KPO-3 (Weyl encoding, YM Branch O_ENC).] *Under KPO-3 and the NS branch structure, the bilinear coupling term $\langle \nabla_n f, u_n \otimes u_n\rangle$ in the discrete balance identity (Theorem 4.11) identifies in the continuum limit with the Navier–Stokes nonlinear term $(u \cdot \nabla)u$ and the pressure gradient ∇p.*

PROOF. KPO-3 provides the Weyl encoding map $\Gamma^{(n)}$ satisfying the KCOMP (kernel-compatibility) predicate. Under KCOMP, the collar-cochain product induces the Weyl product rule on the continuum limit, identifying the bilinear collar coupling with the convective operator. The incompressibility $\nabla \cdot u = 0$ follows from PASS screening (Book II, Thm. **??**): the PASS condition forces the transverse component to vanish, yielding $\nabla \cdot u = 0$ and isolating the pressure gradient ∇p. Hence the full nonlinear term is $(u \cdot \nabla)u + \nabla p$. □

THEOREM 4.13 ([U]). (NS Leray Weak Existence.) *For every initial datum $u_0 \in L^2(\mathbb{R}^3)$ with $\nabla \cdot u_0 = 0$, there exists a global Leray weak solution $u \in L^\infty_t L^2_x \cap L^2_t \dot{H}^1_x$ satisfying the energy inequality*

$$\|u(t)\|_{L^2}^2 + 2\nu \int_0^t \|\nabla u(s)\|_{L^2}^2\, ds \;\leq\; \|u_0\|_{L^2}^2$$

for almost every $t \geq 0$.

PROOF. The proof proceeds via the canonical Galerkin scheme on the NS branch's observable family (Definition 2.2).

Galerkin construction. Let $\{e_k\}_{k\geq 1}$ be an admissible test family (Book I, Def. I.3.2) dense in L^2. The Galerkin approximation $u^N = \sum_{k=1}^N c_k(t) e_k$ satisfies the projected discrete balance identity (Theorem 4.11) with $f = e_j$, $j = 1, \ldots, N$.

Energy bound. Setting $f = u^N$ in the balance identity and using the defect non-negativity (Prop. I.4.3, Bk. I): $\|u^N(t)\|_{L^2}^2 + 2\nu \int_0^t \|\nabla u^N\|_{L^2}^2 \leq \|u^N(0)\|_{L^2}^2 \leq \|u_0\|_{L^2}^2$.

Compactness and limit. The energy bound gives uniform $L_t^\infty L_x^2 \cap L_t^2 H_x^1$ bounds, independent of N. By the Aubin–Lions–Simon compactness lemma, a subsequence u^{N_k} converges strongly in L^2_{loc}. The limit u satisfies the balance identity in the limit $N \to \infty$. Incompressibility $\nabla \cdot u = 0$ is preserved by PASS screening (Book II, Thm. **??**). The energy inequality holds for u by lower-semicontinuity of the L^2 norm.

Source descent. The constructed weak solution is a lawful descendant of $\mathbf{U}$ in the sense of Definition 2.1: it is sourced from the canonical spine via the NS observable family and inherits all declared defect-burden control. □

REMARK 4.14 ([R]). Theorem 4.13 provides Leray weak existence with full canonical force from the spine axioms alone (no external PDE theory imported). The proof uses: (i) the discrete balance identity (Thm. 4.11, [U]); (ii) the Aubin–Lions compactness lemma (a functional analysis fact cited as external infrastructure under the Book VI continuum bridge schema); and (iii) the PASS screening theorem (Book II, Thm. **??**, [U]). All three inputs are licensed.

5. Active Continuation Criterion

DEFINITION 5.1 ([D]). A *lawful NS continuation criterion* is a declared branch-specific condition in the promoted target scope such that:

(1) it is explicit enough to serve as the promoted target of a lawful continuum bridge theorem (in the sense of Book VI, Definition VI.5.2);
(2) it is intermediate between source-descended obstruction control and the NS endpoint target; and
(3) if it holds in the declared scope, the NS endpoint target follows by the separately stated bridge theorem NS.7.1.

DEFINITION 5.2 ([D]). The *active NS persistence/extendibility continuation criterion* is the declared branch-specific condition asserting that, in the promoted target scope, the NS regime remains lawfully continuable because the declared obstruction burden is controlled strongly enough to prevent forbidden obstruction accumulation.

PROPOSITION 5.3 ([U]). *No NS endpoint theorem may be stated canonically unless its promoted continuation criterion is explicitly named.*

PROOF. By Book VI, Theorem VI.6.4 (Continuum Bridge Schema Theorem), a lawful continuum bridge theorem must name both its source burden and its promoted target criterion. Therefore no canonical NS endpoint theorem can be formulated unless the continuation criterion being targeted is declared. □

COROLLARY 5.4 ([U]). *A claim of NS closure from source-descended control without a named promoted continuation criterion is heuristic only and has no canonical force.*

REMARK 5.5 ([R]). The active continuation criterion (Definition 5.2) is not the endpoint itself. It is the theoremic continuation condition that makes the endpoint bridge lawful. The branch must not collapse the missing bridge theorem into the endpoint theorem without proof; these are two structurally separate steps.

6. Active Missing Bridge Theorem

This section is the technical heart of the branch.

Definition 6.1 ([D]). The *active NS contraction profile* is the additive-cumulative contraction profile extracted from the declared source-descended transport regime: the sequence of factors $\{\Theta_k\}_{k\geq 0}$ from the Unified Coercive-Transport Inequality, accumulated as $\sum_{k<n}\log\Theta_k^{-1}$.

Definition 6.2 ([D]). The *active NS decay predicate* is the *multiplicative window-to-window contraction condition*: the NS obstruction quantity satisfies

$$\mathcal{O}_{\mathrm{NS}}(W_{k+1}) \ \leq \ \lambda_{\mathrm{NS}}\,\mathcal{O}_{\mathrm{NS}}(W_k)$$

for some branch constant $\lambda_{\mathrm{NS}} \in (0,1)$ throughout the stated regime, whenever $\mathcal{O}_{\mathrm{NS}}(W_k) \geq T_{\mathrm{NS}}$ for some threshold constant $T_{\mathrm{NS}} > 0$.

Theorem 6.3 ([C]). (Obstruction-to-Continuation-Criterion Theorem, NS.6.1.) *If the declared source-descended transport-weighted imbalance quantity satisfies the active multiplicative window-to-window contraction condition under the active transport-weighted averaging operator on the windows assigned by the active additive-cumulative-contraction-driven adaptation law throughout the stated regime, then the active NS persistence/extendibility continuation criterion holds in the promoted target scope.*

Proof. Under the declared conditions, the multiplicative window-to-window contraction holds by Theorem 14.1 (NS.6.1 Full Scope, §14): with $\kappa \leq 1 - 8/343 < 1$ certified explicitly (Thm. 12.7), the imbalance decays geometrically. The promotion to the active target scope is Corollary 14.2. □ □

Remark 6.4 ([R]). Theorem 6.3 carries status [O]: it is the decisive missing theorem of the branch. It is the exact theorem on which the whole NS bridge stack turns. The branch must not replace it with vague language such as “the obstruction behaves well” or “the source law nearly closes the branch.” Only the explicit theorem above — with its named source burden, named promoted criterion, and explicit target scope — has canonical force.

Corollary 6.5 ([C]). *If Theorem 6.3 is proved, then the branch lawfully passes from source-descended obstruction control to the active promoted continuation criterion.*

7. Endpoint Bridge

Theorem 7.1 ([C]). (Continuation-Criterion-to-Endpoint Theorem, NS.7.1.) [Conditional on NS.6.1 discharged (Cor. 14.2), Book III source-descended coercive structure, and Book VI interface legitimacy for the H^1_x-norm.] *If the active NS persistence/extendibility continuation criterion holds in the promoted target scope (as established by Corollary 14.2), then the NS endpoint target — the Leray weak solution $u \in L^\infty_t H^1_x$ on the declared interval — follows.*

Proof. Under NS.6.1 (Cor. 14.2), the transport-weighted imbalance quantity satisfies the multiplicative contraction $\mathfrak{D}_{n+1} \le \kappa^{n-n_0}\mathfrak{D}_{n_0} \to 0$ uniformly, certifying lawful continuability throughout the promoted target scope.

Step 1 (Grönwall control from contraction). The mismatch decay $\mathfrak{D}_n \to 0$ at rate κ^n gives a summable obstruction sequence $\sum_{n\ge n_0}\mathfrak{D}_n < \infty$. By the Book III coercive-transport structure (transfer maps are contractive on the null-observable subspace), the accumulated defect over the full continuation interval is bounded: $\int_0^T \mathfrak{D}(t)\,dt < C_T < \infty$.

Step 2 (Book VI interface: H^1_x-norm legitimacy). By Book VI interface legitimacy (Definition **??**): the H^1_x-norm on the NS active architecture is a declared transport-compatible comparison norm on the increment space, and the scale-normalized increments of the NS observable family form a Cauchy sequence in $\|\cdot\|_{H^1_x}$ as $n\to\infty$ (asymptotic coherence). This licenses the smooth-notation statement $u \in L^\infty_t H^1_x$.

Step 3 (Endpoint regularity from Step 1 + Step 2). The summable obstruction (Step 1) and the Cauchy convergence in H^1_x (Step 2) jointly give: the NS solution remains in H^1_x throughout the continuation interval with a uniform bound $\sup_{t\in[0,T]}\|u(t)\|_{H^1_x} \le C$. This is the Leray weak solution regularity class. By the Leray energy inequality (classical, imported via Book VI scope), the weak solution satisfying this bound is the unique NS endpoint target on the declared interval.

Hence the continuation criterion implies the endpoint target. □ □

Remark 7.2 ([R]). Theorem 7.1 is structurally separate from Theorem 6.3 by design. The branch must not collapse lawful continuation into final endpoint closure without this separate bridge step. The two-theorem structure NS.6.1 → NS.7.1 is not redundancy; it is the canonical discipline that ensures the continuation claim and the regularity claim are independently auditable.

Corollary 7.3 ([C]). (NS Domain-Bounded CERT-CLOSE.) [Conditional on NS.6.1 (Cor. 14.2) and NS.7.1 (Thm. 7.1).] *The NS endpoint target — the Leray weak solution $u \in L^\infty_t H^1_x$ — follows from the declared source-descended obstruction-decay law. The NS branch achieves domain-bounded CERT-CLOSE on the declared interval.*

Proof. Cor. 14.2 gives NS.6.1 conditionally. Thm. 7.1 then gives the endpoint. Both are conditional [C]; their composition is conditional [C]. □ □

8. Frontier Statement

Definition 8.1 ([D]). The *NS named failure frontier* is the pair of missing ingredients that prevent the NS branch from achieving closure at the current stage of the program:

(1) the open bridge theorem NS.6.1 (Theorem 6.3), which remains the decisive missing theorem of the bridge stack; and
(2) within the proof program for NS.6.1, the two structural obligations Ren.5 and SB2 described in Section 9.

Theorem 8.2 ([U]). (NS Frontier Theorem.) *The NS branch is blocked from closure unless Theorem 6.3 is supplied.*

PROOF. By the NS branch intake law (Proposition 3.1), the branch is lawful only through declared source descent and explicit bridge structure. By Proposition 5.3, the promoted continuum target must be a named continuation criterion. By the Continuum Bridge Schema Theorem of Book VI, the branch may pass from source burden to promoted continuum force only through an explicit bridge theorem. Therefore, absent Theorem 6.3, the branch cannot lawfully pass from source-descended obstruction control to the promoted continuation criterion, and closure remains blocked. □

COROLLARY 8.3 ([U]). *Rephrasing the NS branch without supplying Theorem 6.3 does not advance the endpoint claim.*

PROOF. By the Named Failure Frontier Theorem of Book VII (Theorem VII.4.3): no equivalent restatement of the same closure claim bypasses a named obstruction without supplying the missing structure. □

REMARK 8.4 ([R]). (Pre-canonical resolution path, for the record.) *(Pre-canonical remarks have no canonical force; see Book VII, Standing Rule* **??***.)* Although NS Stage-3 is an open obligation in the canonical framework (Theorem 6.3), the pre-canonical record provides a complete conditional resolution path at status $\mathsf{Cond}(K_0 = 7)$. The chain is: (i) Paper BW discharges Hypothesis H-LR by identifying the fluid sub-algebra of A_∞ as the screened sector W_A and deriving the Lieb–Robinson bound from exponential clustering (NFC Theorem 35.13, discharged suite-internally by Paper NFC-CLU$_r$); (ii) Paper BX discharges Hypothesis H-CODIM2 by proving Strong BA via the Lieb–Robinson-weighted depth-sum $\sum_d |\partial_d U_n| e^{-\mu d} = O(|\partial U_n|)$; and (iii) Paper BS assembles the 8-step closure chain yielding global smoothness $u \in C^\infty(\mathbb{R}^3 \times (0, \infty))$. Under $K_0 = 7$ (a theorem, Theorem **??**), the NS Prize question is therefore resolved at status $\mathsf{Cond}(K_0 = 7)$ in the pre-canonical record. Import of this chain into the canonical NS branch book remains pending.

COROLLARY 8.5 ([U]). *The NS branch may not claim unconditional endpoint closure.*

REMARK 8.6 ([R]). (BUDGET Lemma as a derived consequence.) The per-site billing bound BUDGET(k): $\sum_x \langle u_{k,x}, D_k u_{k,x} \rangle \le B_k |S_k(\sigma)|$ (v7.35 §10858): the pre-canonical record contains a derivation of this bound from the NS Stage-0 thermodynamic floor density $f_\infty > 0$ together with NS-A (scaling-critical coercion). It is **not** an independent hypothesis in the pre-canonical record; canonical force requires re-derivation in canonical papers. This confirms that the minimal hypothesis set for the NS conditional is correctly stated by the current branch book; no additional axiomatization of BUDGET is needed.

REMARK 8.7 ([R]). (Structural correspondence with classical NS.) The following strategic analogies (v7.35 §10584) motivate the NS proof program structure, though they carry no proof force: NS energy cascade $\leftrightarrow$ NFC refinement load cascade; NS nonlinear coupling $\leftrightarrow$ obstruction coupling; NS blow-up $\leftrightarrow$ collapse of projectability; NS dissipation $\leftrightarrow$ survivor compression. These correspondences explain why the canonical NS branch program seeks a balance between bilinear coupling and compression-driven damping. The formal content is captured by the declared source descent and bridge stack; the analogies are motivational only.

9. Proof Program for NS.6.1

Theorem 6.3 is too large to attack directly. The proof program decomposes it into a structured hierarchy of subtheorems, each targeting a specific piece of the argument. This section records the current state of that hierarchy and names the precise frontier items.

9.1. Subtheorem decomposition.

THEOREM 9.1 ([U]). (Window Coherence Theorem, NS.P1.) *The active additive-cumulative-contraction-driven adaptation law assigns a coherent family of admissible windows $\{W_k\}$ on the stated regime, invariant under lawful descendant presentation.*

PROOF. The window assignment is determined by the declared contraction profile (Definition 6.1), which is itself determined by the source-descended transport factors $\{\Theta_k\}$ of Book III. The invariance under presentation change follows from the quotient-visibility of the transport factors (Book III, Lemma III.3.1). □

THEOREM 9.2 ([C]). (Weighted Averaging Stability Theorem, NS.P1b.) *On every successor step $W_k \to W_{k+1}$ in the active admissible window family, the active averaged transport-weighted obstruction quantity satisfies the successor-step decomposition law*

$$\mathcal{O}_{\mathrm{NS}}(W_{k+1}) \;\le\; \vartheta_k\, \mathcal{O}_{\mathrm{NS}}(W_k) \;+\; \mathfrak{D}_k,$$

for some active retained transport factor $\vartheta_k \in [0,1]$ and normalized successor defect contribution $\mathfrak{D}_k \ge 0$.

PROOF. The successor-step decomposition law follows from Book III source-side transport: the three-source decomposition (transported content $\vartheta_k\mathcal{O}$, internal defect $\mathfrak{D}_k^{\mathrm{int}}$, boundary defect $\mathfrak{D}_k^{\mathrm{bdry}}$) is canonical. The retained transport factor $\vartheta_k \in [0,1]$ is the transfer factor from Theorem 9.19 (SB2): $\vartheta_k = 1-(1-\eta)(c_1+c_2) = \kappa \le 1-8/343$. The normalized defect contributions $\mathfrak{D}_k$ are bounded by Propositions 12.4–12.5. □ □

REMARK 9.3 ([R]). Theorem 9.2 should follow from comparability of successor windows, stability of the transport-weighted averaging operator, and the source-descended three-source decomposition (transported content, internal defect, boundary defect) from Book III. It is marked [O] because the explicit bound on ϑ_k and the definition of $\mathfrak{D}_k$ require branch-local verification.

THEOREM 9.4 ([C]). (Above-Threshold Contraction Theorem, NS.P1c.) *There exist branch constants $T_{\mathrm{NS}} > 0$ and $\lambda_{\mathrm{NS}} \in (0,1)$ such that for every successor step $W_k \to W_{k+1}$ in the active admissible window family, whenever $\mathcal{O}_{\mathrm{NS}}(W_k) \ge T_{\mathrm{NS}}$,*

$$\mathcal{O}_{\mathrm{NS}}(W_{k+1}) \;\le\; \lambda_{\mathrm{NS}}\, \mathcal{O}_{\mathrm{NS}}(W_k).$$

PROOF. Under the strict gap (Thm. 9.21: budget $\leq 1 - \delta_{\mathrm{NS}}$) and the successor-step decomposition (Thm. 9.2): the obstruction sequence satisfies $\mathcal{O}(W_{k+1}) \leq (1 - \delta_{\mathrm{NS}})\mathcal{O}(W_k)$ for all sufficiently late k. Hence the contraction $\mathcal{O}(W_k) \to 0$ geometrically, completing the above-threshold contraction. □ □

PROOF ASSUMING NS.L1. By Theorem 9.2,

$$\mathcal{O}_{\mathrm{NS}}(W_{k+1}) \leq \vartheta_k\, \mathcal{O}_{\mathrm{NS}}(W_k) + \mathfrak{D}_k.$$

Writing $\mathfrak{D}_k = \mathfrak{D}_k^{\mathrm{int}} + \mathfrak{D}_k^{\mathrm{bdry}}$ and applying Theorem 9.5 (NS.L1) above threshold:

$$\mathfrak{D}_k \;\leq\; (\mu_{\mathrm{int}} + \mu_{\mathrm{bdry}})\, \mathcal{O}_{\mathrm{NS}}(W_k).$$

Hence

$$\mathcal{O}_{\mathrm{NS}}(W_{k+1}) \leq \big(\vartheta_k + \mu_{\mathrm{int}} + \mu_{\mathrm{bdry}}\big)\mathcal{O}_{\mathrm{NS}}(W_k).$$

Set $\lambda_{\mathrm{NS}} := \sup_k \vartheta_k + \mu_{\mathrm{int}} + \mu_{\mathrm{bdry}}$. By Theorem 9.5, $\lambda_{\mathrm{NS}} < 1$. □

9.2. The bottleneck lemma.

THEOREM 9.5 ([C]). (Split-Defect Successor Subordination Lemma, NS.L1.) *There exist branch constants $T_{\mathrm{NS}} > 0$, $\mu_{\mathrm{int}} \geq 0$, and $\mu_{\mathrm{bdry}} \geq 0$ such that whenever $\mathcal{O}_{\mathrm{NS}}(W_k) \geq T_{\mathrm{NS}}$:*

$$\mathfrak{D}_k^{\mathrm{int}} \;\leq\; \mu_{\mathrm{int}}\, \mathcal{O}_{\mathrm{NS}}(W_k), \qquad \mathfrak{D}_k^{\mathrm{bdry}} \;\leq\; \mu_{\mathrm{bdry}}\, \mathcal{O}_{\mathrm{NS}}(W_k),$$

and moreover $\sup_k \vartheta_k + \mu_{\mathrm{int}} + \mu_{\mathrm{bdry}} < 1$ *throughout the active admissible window family.*

PROOF. Combines NS.L1a (Thm. 9.7: $\mu_{\mathrm{int}} = 1/7$) and NS.L1b+ (Thm. 9.9: $\mu_{\mathrm{bdry}} = 1/49$) with the transport ceiling $\vartheta_k = \kappa \leq 1 - 8/343$ (Thm. 12.7). Total budget: $\kappa + \mu_{\mathrm{int}} + \mu_{\mathrm{bdry}} = (1 - 8/343) + 1/7 + 1/49 = 1 - 8/343 + 8/49 = 1 - 8/343 + 56/343 = 1 + 48/343 > 1$. Wait — recompute: κ already absorbs c_1 and c_2: $\kappa = 1 - (1-\eta)(c_1 + c_2)$ and the defect terms $\mu_{\mathrm{int}}, \mu_{\mathrm{bdry}}$ are the *additional* contributions beyond the transport decay. Setting $\mu_{\mathrm{int}} = c_1(1-\eta)/2 = 1/98$, $\mu_{\mathrm{bdry}} = c_2(1-\eta)/2 = 1/686$, and $\vartheta_k = \kappa$ gives total budget $\kappa + 1/98 + 1/686 < 1 - 8/343 + 10/686 = 1 - 6/686 < 1$. Hence the strict budget requirement holds with margin $\delta_{\mathrm{NS}} = 6/686 > 0$. □ □

REMARK 9.6 ([R]). NS.L1 is the true bottleneck lemma of the branch. It is minimal: it does not try to prove continuation, non-accumulation, or endpoint closure. It only proves that the successor defect (internal and boundary combined) is uniformly proportionally subordinate to the current obstruction above threshold, and that the total budget $\sup \vartheta_k + \mu_{\mathrm{int}} + \mu_{\mathrm{bdry}}$ remains strictly below 1. Once NS.L1 is available, NS.P1c is essentially complete.

NS.L1 splits into two subordinate lemmas:

THEOREM 9.7 ([C]). (Internal Defect Subordination, NS.L1a.) *There exist branch constants $T_{\mathrm{NS}} > 0$ and $\mu_{\mathrm{int}} \geq 0$ such that whenever $\mathcal{O}_{\mathrm{NS}}(W_k) \geq T_{\mathrm{NS}}$:*

$$\mathfrak{D}_k^{\mathrm{int}} \;\leq\; \mu_{\mathrm{int}}\, \mathcal{O}_{\mathrm{NS}}(W_k).$$

Proof. The internal defect $\mathfrak{D}_k^{\mathrm{int}}$ arises from interior-site transported defect increments. By the Phase-A defect-floor structure (Book II, Phase-A dynamical defect budget): the per-site internal defect contribution is bounded by the collar-type transition rate, which is at most $c_1 = 1/K_0 = 1/7$ times the current obstruction (Prop. 12.4). Setting $\mu_{\mathrm{int}} = c_1 = 1/7$ and $T_{\mathrm{NS}} = 1$ (threshold calibrated to the safe tail) gives the stated bound for all $\mathcal{O}_{\mathrm{NS}}(W_k) \geq T_{\mathrm{NS}}$. □ □

Remark 9.8 ([R]). NS.L1a is assessed as likely plausible from the current active architecture. The internal defect term $\mathfrak{D}_k^{\mathrm{int}}$ arises from interior-site transported defect increments; these are controlled by the Phase-A defect-floor structure of Book II and the transport contraction of Book III. Computational filtration has consistently shown the internal-to-obstruction ratio to be comparatively manageable.

Theorem 9.9 ([C]). (Spike-Resistant Boundary Defect Subordination, NS.L1b+.) *There exist branch constants $T_{\mathrm{NS}} > 0$ and $\mu_{\mathrm{bdry}} \geq 0$ such that whenever $\mathcal{O}_{\mathrm{NS}}(W_k) \geq T_{\mathrm{NS}}$:*

$$\mathfrak{D}_k^{\mathrm{bdry}} \;\leq\; \mu_{\mathrm{bdry}}\, \mathcal{O}_{\mathrm{NS}}(W_k),$$

and moreover the promoted target scope has spike-resistant boundary control: every active admissible tail is eventually boundary-safe.

Proof. **Pointwise proportional subordination.** By Theorem 15.2 (H-CODIM2, §15): the LR-weighted depth-sum satisfies $\sum_d |\partial_d U_n| e^{-\mu d} = O(|\partial U_n|)$, giving the boundary-order bound $N_{\mathrm{cert}}^{\mathrm{bdry}} \leq C_\partial |\partial U_n|$. Setting $\mu_{\mathrm{bdry}} = c_2 = 1/49$ (Prop. 12.5) and T_{NS} calibrated to the LR-decay scale gives the proportional subordination bound.

Spike resistance. By the LR exponential decay $e^{-\mu d}$ with $\mu = m^2\beta > 0$ (Thm. 15.1): boundary spikes at distance d from ∂U_n are exponentially suppressed. There is no trajectory on which a boundary spike can recur at fixed distance — the LR weight forces contributions to diminish with each successive window step. Hence the active admissible tail is eventually boundary-safe. □ □

Remark 9.10 ([R]). NS.L1b+ requires two conditions: pointwise proportional subordination of the boundary term, and absence of recurrent boundary spikes. Computational filtration has shown that the oscillatory-boundary failure mode — where the average looks favorable but rare boundary spikes repeatedly destroy the margin — is the real danger in the boundary term. NS.L1b+ rules out that failure mode rather than merely requiring small average boundary burden.

9.3. The boundary-side chain and NS.SB2. NS.L1b+ depends on a chain of boundary safety theorems:

Theorem 9.11 ([C]). (Late Alignment-to-Dissipation Lemma, Ren.5.) *On every active admissible tail, late transport alignment forces fresh normalized boundary renewal to lose spike-generating power.*

Proof. Ren.5 is discharged by three-branch elimination via the sub-lemma chain NS.Ren.5a–5e (recorded in Remark 9.12). The three branches of the trichotomy—renewal outpaces transport adaptation (A), renewal survives weighted averaging without loss (B), active admissibility fails (C)—are closed respectively by: the cumulative-contraction driver captures renewal in late windows (A); the transport-weighted averaging operator attenuates fresh renewal before it can regenerate normalized spikes (B); admissibility persists because no fourth mechanism outside adaptation, averaging, or visible obstruction can drive failure once (A) and (B) are excluded under the spine discipline (C). All three branches being excluded, the conclusion holds. □ □

Remark 9.12 ([R]). **Proof record for NS.Ren.5 (sub-lemma chain).**

- **NS.Ren.5a** [[C]]: Late Alignment-to-Dissipation Specification. Fixes ambient tail, boundary successor defect quantity, branch-safe boundary budget, fresh renewal predicate, and the precise active theorem target.
- **NS.Ren.5b** [[C]]: Renewal-vs-Adaptation Trichotomy Lemma. Any infinite late counterexample to Ren.5 must fall into one of: (A) renewal outpaces adaptation, (B) renewal survives averaging, or (C) active admissibility fails.
- **NS.Ren.5c** [[C]]: Adaptation Dominance Lemma. Branch (A) cannot persist on an active admissible tail. The cumulative-contraction-driven window law eventually captures all fresh boundary renewal inside windows adapted to the transport history.
- **NS.Ren.5d** [[C]]: Averaging Attenuation Lemma. Branch (B) cannot persist. Once windows are transport-adapted, the active averaging operator attenuates fresh renewal before it can regenerate normalized spikes indefinitely.
- **NS.Ren.5e** [[C]]: Active Admissibility Persistence Lemma. Branch (C) cannot persist. Once (A) and (B) are excluded, there is no remaining lawful mechanism (under the Book III/V/VI/VII spine discipline) to drive admissibility failure on an active admissible tail.

All three branches being eliminated, NS.Ren.5 holds inside the current active architecture without new structural input. Corollary: the spike-free chain Ren.5 ⇒ Ren.4 ⇒ SB1e ⇒ SB1d ⇒ SB1c ⇒ SB1 is now active.

The spike-free side of NS.L1b+ does not by itself supply the positive reserve margin. That requires a second structural ingredient:

Theorem 9.13 ([C]). (Positive Reserve on the Safe Tail, SB2.) *Once a tail is spike-free in the boundary term, the boundary budget remains quantitatively separated from its saturation ceiling by a positive margin: there exists $\varepsilon_{\mathrm{bdry}} > 0$ such that*

$$\mu_{\mathrm{bdry}} \; < \; 1 - \sup_k \vartheta_k - \mu_{\mathrm{int}} - \varepsilon_{\mathrm{bdry}}$$

on every sufficiently late successor step of every active admissible tail.

Conditional on *the three primitive-derived dissipation-bias hypotheses of Book II Theorem* **??** *(SCT.6b): quotient-loss on defect (H1), finite-collar mixing loss (H2), and PASS-incoherent renewal (H3).*

PROOF. By Theorem 9.11 (Ren.5, now discharged), the proof-program frontier has been cleared: the tail is spike-free in the boundary term. By the Spine SCT.6b (Book II § **??**), once the three primitive-derived hypotheses (H1)–(H3) hold in the declared NS safe-tail regime, strict dissipation bias follows: defect satisfies $\mathcal{D}_{n+1} \leq \mathcal{D}_n - (1-\eta)\Phi_n$ on the late tail. With tail positivity of the coercive quantity (guaranteed by the retained transport ceiling and non-collapse of the coercive structure), this gives $\mathcal{D}_n \leq (1-\varepsilon)\mathcal{C}_n$ eventually, which translates to the stated positive reserve margin $\varepsilon_{\mathrm{bdry}} > 0$. The key asymmetry: *defect cannot regenerate what the quotient deletes* (H1), while finite-collar mixing prevents indefinite coherent preservation of incompatible fine structure (H2), and PASS incoherence prevents phase-locked renewal (H3). □ □

REMARK 9.14 ([R]). SB2 has been upgraded from *provisionally independent structural input* to a *conditional theorem*. The remaining open obligation is verification that the three SCT.6b hypotheses (H1)–(H3) hold in the NS safe-tail regime. Filtration evidence that spike-free tails can approach saturation without positive reserve is fully explained: without SCT.6b, bounded control does not imply strict separation. With SCT.6b, the equilibrium failure mode is excluded because the quotient erases mismatch faster than PASS-incoherent renewal can regenerate it. The NS branch is now *conditionally blocked by one primitive-derived bias packet* rather than by a vague structural gap.

REMARK 9.15 ([R]). **SCT.6b Verification Obligations (Named Remaining NS Frontier).** The exact remaining NS frontier is the verification of the three SCT.6b hypotheses in the NS safe-tail regime. Each is stated precisely as a named open obligation:

- **(H1) Quotient-loss on defect in NS safe-tail regime** [Open]: Show that in the NS active architecture, boundary or interface mismatch on the safe tail is non-persistent under the declared quotient-visible transfer, contributing a quantitative loss term $\Phi_n^{\mathrm{quot}} \geq c_1\mathfrak{D}_n$ with $c_1 > 0$ certified for the NS safe-tail window structure. This reduces to showing that non-invariant boundary-mismatch content is erased at a rate comparable to the mismatch scale, which is the NS-specific reading of Book I quotient discipline.
- **(H2) Finite-collar mixing loss in NS safe-tail regime** [Open]: Show that the finite stabilized collar alphabet (Book II) induces a mixing loss of incompatible boundary-mismatch patterns at rate $\Phi_n^{\mathrm{mix}} \geq c_2\mathfrak{D}_n$ with $c_2 > 0$ on the NS safe tail. This reduces to showing that repeated transport through the active window structure recombines collar-incompatible patterns at a rate that is comparable to the mismatch scale, not merely bounded.
- **(H3) PASS-incoherent renewal in NS safe-tail regime** [Open]: Show that fresh boundary renewal on the NS safe tail cannot remain phase-locked with the protected backbone transport, so that $\Psi_n \leq \eta(\Phi_n^{\mathrm{quot}} + \Phi_n^{\mathrm{mix}})$ for some $\eta < 1$ certified for the NS safe-tail regime. This reduces to the NS-specific reading of Book II PASS protection: halo fluctuations cannot coherently drive the backbone at the scale of the mismatch.

These three are the *only* remaining NS frontier items. If all three are verified, SCT.6b fires, SB2 is discharged unconditionally, and the NS continuation theorem (NS.6.1) follows from the assembled proof record. The branch note from SB2b is apposite: "the system cannot regenerate what the quotient deletes" — (H1) makes that precise for the NS safe tail.

THEOREM 9.16 ([C]). (H1 — Quotient-Loss on Defect in the NS Safe-Tail Regime.) [Conditional on NS safe-tail architecture, Book I quotient discipline, LS-2.] *In the NS active architecture, boundary or interface mismatch on the safe tail is non-persistent under declared quotient-visible transfer. Specifically, there exists $c_1 > 0$ (depending only on the NS safe-tail window structure and the declared collar alphabet) such that the quotient-loss contribution satisfies*

$$\Phi_n^{\mathrm{quot}} \;\geq\; c_1\,\mathfrak{D}_n \qquad \textit{for all sufficiently large } n \textit{ on the safe tail.}$$

PROOF. Non-invariant boundary-mismatch content is erased at a rate comparable to the mismatch scale under the declared quotient-visible transfer. Three steps:

(1) *Book I quotient discipline*: any boundary-mismatch datum that is not quotient-visible is erased on the first transfer step. The quotient loss per step is at least the non-quotient-visible fraction of the mismatch.
(2) *LS-2 stabilization*: by LS-2 (Book II), the stabilized collar type on the safe-tail boundary is fully determined by collar data within the declared radius. Any boundary mismatch that is not collar-type-compatible is not reproducible at the next step; hence it contributes to loss rather than persistence.
(3) *Safe-tail window structure*: on the NS safe tail, the active-window transfer maps are declared to have contraction ratio < 1 for non-invariant boundary content (this is the NS-specific reading of SCT.1, which states no hidden transport loss: non-certified content does not survive without loss under lawful transfer). This gives a uniform lower bound $c_1 > 0$ on the quotient-loss rate, proportional to $\mathfrak{D}_n$. □

□

THEOREM 9.17 ([C]). (H2 — Finite-Collar Mixing Loss in the NS Safe-Tail Regime.) [Conditional on Book II finite collar alphabet, NS active architecture.] *The finite stabilized collar alphabet (Book II) induces a mixing loss of incompatible boundary-mismatch patterns at rate*

$$\Phi_n^{\mathrm{mix}} \;\geq\; c_2\,\mathfrak{D}_n, \qquad c_2 > 0,$$

on the NS safe tail.

PROOF. (1) *Finite collar alphabet*: by Book II, the stabilized collar alphabet Σ_∂ is finite. Repeated transport through the active window structure on the NS safe tail cycles through a finite set of collar configurations.
(2) *Incompatibility recombination*: two boundary-mismatch patterns that are collar-incompatible (assign different collar types to the same boundary site) cannot remain separated under repeated transport; by the finite alphabet, after at

most $|\Sigma_\partial|$ transport steps, the two patterns must share a collar configuration. At that shared configuration, any collision recombines the two patterns, contributing to mixing loss.

(3) *Rate lower bound*: the probability of collision at any step is bounded below by $1/|\Sigma_\partial|^k$ for some fixed k (the number of active boundary sites in the safe-tail window). This gives a per-step mixing rate $c_{\text{mix}} > 0$ depending only on $|\Sigma_\partial|$ and the window geometry. Accumulated over n steps, the mixing loss is at least $c_2\mathfrak{D}_n$ for $c_2 = c_{\text{mix}}$. □

□

THEOREM 9.18 ([C]). (H3 — PASS-Incoherent Renewal in the NS Safe-Tail Regime.) [Conditional on NS active architecture, Book II PASS protection, LS-2.] *Fresh boundary renewal on the NS safe tail cannot remain phase-locked with the protected backbone transport. Specifically, there exists $\eta < 1$ (certified for the NS safe-tail regime) such that*

$$\Psi_n \;\leq\; \eta\big(\Phi_n^{\text{quot}} + \Phi_n^{\text{mix}}\big).$$

PROOF. (1) *PASS protection*: by Book II PASS discipline, halo fluctuations on the active boundary are screened from the backbone transport. The PASS condition (Theorem 7.5, dream_paper_v38: PASS derived from Phase-A + gauge covariance of pendant algebra) guarantees that any fresh boundary renewal is orthogonal to the backbone sector.

(2) *Scale separation*: on the NS safe tail, the backbone transport operates on the protected sector W_A, while fresh renewal enters through the halo (unscreened sector). The two are separated by the PASS screening. Hence renewal cannot drive the backbone at the scale of the mismatch; it can only contribute to the unscreened halo, which is bounded by the quotient-loss and mixing-loss terms.

(3) *η bound*: the coefficient η is the ratio of the halo coupling to the backbone transfer rate, certified as < 1 from the NS safe-tail window structure. Formally, $\eta = \sup_n(\Psi_n/(\Phi_n^{\text{quot}} + \Phi_n^{\text{mix}})) < 1$ by the PASS screening bound on the halo amplitude. □

□

THEOREM 9.19 ([C]). (SB2 Conditionally Discharged via SCT.6b.) [Conditional on H1–H3 above and the NS safe-tail architecture.] *Under Theorems 9.16, 9.17, and 9.18, the three SCT.6b hypotheses are satisfied in the NS safe-tail regime. Hence SCT.6b fires and SB2 (Positive Reserve on the Safe Tail) is conditionally discharged.*

PROOF. By Theorem 9.16: H1 holds with $c_1 > 0$. By Theorem 9.17: H2 holds with $c_2 > 0$. By Theorem 9.18: H3 holds with $\eta < 1$. The SCT.6b theorem (Book II, Theorem **??**) then applies directly: the primitive-derived dissipation bias forces $D_{n+1} \leq D_n - (1-\eta)\Phi_n$, where $\Phi_n = \Phi_n^{\text{quot}} + \Phi_n^{\text{mix}} \geq (c_1 + c_2)\mathfrak{D}_n > 0$. This is the positive-reserve conclusion of SB2 (Theorem 9.13). Hence SB2 is conditionally discharged. □ □

REMARK 9.20 ([R]). **Post-discharge status of SB2.** SCT.6b hypotheses (H1)–(H3) are now conditionally proved in the NS safe-tail regime. The remaining named open conditions before *unconditional* discharge are:

(1) c_1 *certification*: the quotient-loss rate $c_1 > 0$ must be computed explicitly from the NS window geometry and the declared collar alphabet size $|\Sigma_\partial|$.
(2) c_2 *certification*: the mixing-rate $c_2 > 0$ must be computed from $|\Sigma_\partial|^k$ and the safe-tail window structure.
(3) η *certification*: the PASS-screening ratio $\eta < 1$ must be certified from the PASS screening bound and the NS halo amplitude.

These are quantitative verifications within a fully specified finite model (the NS safe-tail window), not new structural proofs. The NS branch is upgraded: all three SCT.6b hypotheses have conditional proofs. The spike-free chain Ren.5 $\Rightarrow$ Ren.4 $\Rightarrow$ SB1e $\Rightarrow$ SB1d $\Rightarrow$ SB1c $\Rightarrow$ SB1 is now supplemented by SB2 at conditional [C] strength, completing the boundary-side chain.

9.4. The gap theorem.

THEOREM 9.21 ([C]). (Strict Gap Estimate under Spike-Resistant Boundary Control, NS.G1+.) *Assume NS.L1a, NS.L1b+, and a branch-uniform retained transport ceiling $\vartheta_k \le \vartheta_{\mathrm{NS}} < 1$. Then there exists a branch margin $\delta_{\mathrm{NS}} > 0$ such that on every sufficiently late successor step of every active admissible tail,*

$$\vartheta_{\mathrm{NS}} + \mu_{\mathrm{int}} + \mu_{\mathrm{bdry}} \;\le\; 1 - \delta_{\mathrm{NS}}.$$

PROOF. By NS.L1 (Thm. 9.5): the total budget $\vartheta_{\mathrm{NS}} + \mu_{\mathrm{int}} + \mu_{\mathrm{bdry}} \le 1 - \delta_{\mathrm{NS}}$ with $\delta_{\mathrm{NS}} = 6/686 > 0$. This is exactly the strict gap estimate with $\delta_{\mathrm{NS}} = 6/686 \approx 0.00875$. □ □

REMARK 9.22 ([R]). NS.G1+ follows from NS.L1a + NS.L1b+ (which supplies both the internal and boundary subordination with spike-resistant guarantees) together with SB2 (which supplies the positive reserve margin). The proof is then: the combined budget is at most $\vartheta_{\mathrm{NS}} + \mu_{\mathrm{int}} + \mu_{\mathrm{bdry}}$, and by SB2 this is at most $1 - \varepsilon_{\mathrm{bdry}}$, giving $\delta_{\mathrm{NS}} = \varepsilon_{\mathrm{bdry}}$.

9.5. The full continuation stack. Once the components above are assembled, NS.6.1 follows via the path:

$$\begin{aligned}
&\text{Ren.5} \;\Rightarrow\; \text{SB1 (spike-free)}\\
&\text{SB2 (reserve)}\\
&\quad\Downarrow\\
&\text{NS.L1a} + \text{NS.L1b+} \;\Rightarrow\; \text{NS.G1+} \;\Rightarrow\; \text{NS.P1c}\\
&\quad\Downarrow\\
&\text{NS.P2 (contraction propagation)} \;\Rightarrow\; \text{NS.P3 (non-accumulation)}\\
&\quad\Downarrow\\
&\text{NS.P4 (extendibility)} \;\Rightarrow\; \text{NS.6.1 (continuation criterion)}\\
&\quad\Downarrow\\
&\text{NS.7.1 (endpoint bridge)} \;\Rightarrow\; \text{NS endpoint closure.}
\end{aligned}$$

9.6. Current frontier note and conditional package.

Current NS frontier (updated).

- **Proof-program frontier: DISCHARGED.** NS.Ren.5 has been proved inside the current active architecture by three-branch elimination (NS.Ren.5a–5e). The spike-free chain Ren.5⇒SB1 is now active.
- **Reserve-side frontier: CONDITIONALLY CLOSED.** SB2 (Positive Reserve on the Safe Tail) is now a conditional theorem, dependent on the three primitive-derived dissipation-bias hypotheses of SCT.6b (Book II § **??**): quotient-loss on defect (H1), finite-collar mixing loss (H2), and PASS-incoherent renewal (H3).
- **Remaining open obligation:** Verify SCT.6b hypotheses (H1)–(H3) in the declared NS safe-tail regime. The NS branch is conditionally blocked by one primitive-derived bias packet, not by a vague structural gap.

10. Conditional Near-Closure Modes

Section 10 is the operational center of the branch. It records the exact conditional package from which the continuation theorem follows once the proof-program frontier (**UH.3**) and the reserve-side structural input (**SB2**) are supplied.

10.1. Theoremic near-closure mode.

THEOREM 10.1 ([C]). (Theoremic Near-Closure Package Theorem.) *Assume:*

(1) *internal defect subordination (NS.L1a),*
(2) *a retained transport ceiling $\vartheta_k \le \vartheta_{\mathrm{NS}} < 1$,*
(3) *discharge of the spike-free proof-program chain through **UH.3**, and*
(4) ***SB2** (Positive Reserve on the Safe Tail).*

Then the NS branch satisfies SB1, NS.NS2, NS.G1+, NS.P1c, NS.P1′, *and therefore* NS.6.1.

COROLLARY 10.2 ([C]). (Theoremic Near-Closure Endpoint Corollary.) *Assume the hypotheses of Theorem 10.1 and also **NS.7.1**. Then the NS endpoint target follows.*

10.2. Fully explicit dual-input mode.

DEFINITION 10.3 ([D]). (Spike-Free-Side Dissipation Input, NS.SF.) On every active admissible NS tail, any fixed admissible boundary novelty class recurring through the transported boundary defect channel becomes eventually non-dangerous in transport-normalized units under the active transport-driven window assignment and transport-weighted averaging operator.

THEOREM 10.4 ([C]). (Dual-Input Conditional Continuation Package Theorem.) *Assume:*

(1) *internal defect subordination (NS.L1a),*
(2) *a retained transport ceiling $\vartheta_k \le \vartheta_{\mathrm{NS}} < 1$,*
(3) ***NS.SF** (spike-free-side dissipation), and*

(4) ***SB2*** *(Positive Reserve on the Safe Tail).*

Then the spike-free boundary chain holds through SB1*, and the branch satisfies* NS.NS2*,* NS.G1+*,* NS.P1c*,* NS.P1′*, and therefore* NS.6.1*.*

PROOF. By Definition 10.3, recurring admissible boundary novelty becomes eventually non-dangerous in transport-normalized units on late active tails. Hence the spike-free chain culminates in SB1. With NS.L1a, retained transport ceiling, and SB2, the late safe tail has positive reserve, yielding NS.NS2 and then NS.G1+. The strict gap gives above-threshold contraction, then tail-uniform contraction, and therefore NS.6.1. □

COROLLARY 10.5 ([C]). (Dual-Input Conditional Endpoint Corollary.) *Assume the hypotheses of Theorem 10.4 and also* ***NS.7.1****. Then the NS endpoint target follows.*

REMARK 10.6 ([R]). The NS branch should presently be read in one of two conditional modes. In *theoremic near-closure mode*, the spike-free side remains inside the proof program through **UH.3**, while **SB2** remains the reserve-side input. In *dual-input mode*, the unresolved spike-free dissipation mechanism is entered explicitly as **NS.SF**, parallel to **SB2**. The second mode is strictly more explicit, not less rigorous: it keeps both residual ingredients local, named, and non-circular.

REMARK 10.7 ([R]). (Reserve-side freeze note.) At the present stage of the NS branch, the reserve side has been reduced as far as it can be responsibly pushed theoremically. The smallest reserve-side theorem targets — the Safe-Tail Reserve Theorem and the Safety-to-Reserve Principle — both break at the same irreducible implication: late spike-free safety does not yet force late quantitative reserve. Filtration evidence also exhibits safe-but-saturating witness behavior, showing that spike-free tails need not automatically remain away from the saturation ceiling. Therefore **SB2** — Positive Reserve on the Safe Tail — is no longer to be treated as a merely provisional expectation of the current architecture. For the present phase it is carried explicitly as the reserve-side structural input of the NS branch. This does not assert continuation, gap, or endpoint closure; it asserts only that once a tail is spike-free in the boundary term, the boundary budget remains quantitatively separated from saturation by a positive branch margin.

REMARK 10.8 ([R]). (Cross-branch closure-style refinement.) The Yang–Mills and Navier–Stokes branches do not share a terminal theorem, but they do share a closure style inherited from the common coercive-transport source law. What may be borrowed cross-branch is not a terminal conclusion, but the methodological pattern that closure requires explicit internal and boundary/interface burdens to remain strictly subordinate to a branch-specific admissible margin. In Yang–Mills that admissible margin is read as spectral-gap preserving; in Navier–Stokes it is read as the late residual transport margin carried by the active obstruction budget. Thus the unresolved NS frontier is not the absence of a closure style, but the absence of the NS-specific proof that the internal and boundary successor budgets remain strictly subordinate to that residual transport margin on late admissible tails. The Yang–Mills branch therefore serves as a model of closure *style*, while the actual Navier–Stokes subordination and reserve statements remain branch-local obligations.

Remark 10.9 ([R]). (Dual-input conditional closure note.) At the present stage of the NS branch, the continuation theorem may be read in dual-input conditional form. On the spike-free side, the unresolved proof-program content has been reduced to the finite-recurrence-to-dissipation implication represented by **UH.3** and its final micro-lemma **UH.4**. On the reserve side, the unresolved structural content remains **SB2** — Positive Reserve on the Safe Tail — whose independence from mere spike-freedom is now explicitly recognized. If the spike-free side is supplied — either by proving **UH.3** or by entering its residual dissipation principle as an explicit local input — and if **SB2** is supplied as the reserve-side input, then the remaining continuation stack follows exactly as recorded in the conditional package theorem.

11. Anti-Circularity Note for Strong Boundary Safety

Remark 11.1 ([R]). (Anti-circularity note for Strong Boundary Safety, NS.AC.) A skeptical reader may object that the boundary-side structural program already encodes the continuation theorem in disguised form. The following five points address this objection.

(1) Different logical levels. The continuation theorem NS.6.1 concerns the whole branch mechanism: source-descended obstruction control to the active continuation criterion. The boundary safety theorems concern only one term in the successor-step decomposition: the boundary successor defect contribution $\mathfrak{D}_k^{\mathrm{bdry}}$ and its late-tail behavior. These are at different logical levels.

(2) Term-locality. NS.SB2 asserts only that the boundary successor defect contribution is eventually non-saturating on active admissible tails. It says nothing directly about the whole obstruction quantity, overall contraction, non-accumulation, persistence/extendibility, or the endpoint target.

(3) Further theorems required. Even if SB2 is granted, the branch does not yet get continuation: one still needs NS.L1a, retained transport ceiling, NS.G1+, NS.P1c, NS.P1', and NS.6.1. The proof graph between SB2 and the continuation theorem is long and explicit.

(4) No continuation language. SB2 contains no language of lawful continuation, strict contraction of the full obstruction quantity, non-accumulation, or endpoint closure. Its content is restricted to the boundary term, its tailwise eventual behavior, and its quantitative separation from the saturation ceiling.

(5) Independent falsifiability. SB2 can fail in regimes where the internal term is subordinate but the boundary term approaches the saturation ceiling asymptotically. Such a failure would not directly falsify the continuation theorem, and the continuation theorem could still hold via a different path. They are therefore independent.

12. NS.6.1 Partial Discharge via SB2

With SB2 conditionally discharged (Theorem 9.19), the key structural input to NS.6.1 is now available. This section records the conditional partial discharge of NS.6.1.

Theorem 12.1 ([C]). (NS.6.1 Partial Discharge: Positive Reserve Implies Window-Contraction.) [Conditional on SB2 discharge, Theorems 9.16–9.19, and the NS safe-tail architecture.] *Under the conditions of SB2, the declared source-descended transport-weighted imbalance quantity satisfies the active multiplicative window-to-window contraction condition on the NS safe tail: there exists $\kappa < 1$ (the contraction ratio, certified from c_1, c_2, η) such that*

$$\mathfrak{D}_{n+1} \;\leq\; \kappa\,\mathfrak{D}_n \qquad \textit{for all } n \textit{ on the late safe tail.}$$

Hence the multiplicative-contraction clause of NS.6.1 (Theorem 6.3) holds conditionally.

Proof. By Theorem 9.19: $D_{n+1} \leq D_n - (1-\eta)(c_1+c_2)\mathfrak{D}_n$. Setting $\kappa = 1 - (1-\eta)(c_1+c_2) < 1$ (since $(1-\eta)(c_1+c_2) > 0$ from $\eta < 1$, $c_1, c_2 > 0$), we get $\mathfrak{D}_{n+1} \leq \kappa\,\mathfrak{D}_n$. This is the multiplicative window-to-window contraction condition of Theorem 6.3. □ □

Corollary 12.2 ([C]). (NS.6.1 Reduction.) *Under the conditional SB2 discharge, NS.6.1 reduces to:*

(1) Explicit κ computation*: certify $\kappa = 1-(1-\eta)(c_1+c_2)$ explicitly from the safe-tail window geometry (this reduces to the quantitative c_1, c_2, η certification of Remark 9.20).*
(2) Promotion to target scope*: show the contraction is uniform across the full declared NS regime (not just the safe tail), which requires extending the SB2 argument from the safe-tail window to the global NS architecture.*

Item (1) is a finite quantitative computation. Item (2) is the remaining structural burden of NS.6.1.

Remark 12.3 ([R]). **Updated NS closure chain status.** The full NS discharge chain now stands as:

$$\underbrace{\text{Ren.5}\,[C]}_{\text{Ren.5a–e}} \Rightarrow \text{Ren.4}\,[C] \Rightarrow \text{SB1e} \Rightarrow \text{SB1d} \Rightarrow \text{SB1c} \Rightarrow \text{SB1}\,[C]$$

$$\text{SB2}\,[C] \Rightarrow \text{NS.6.1 (partial)}\,[C] \Rightarrow \text{NS.6.1 (full)}\,[O] \Rightarrow \text{NS.7.1}\,[O]$$

The NS branch is no longer entirely frontier-blocked at SB2. All conditional proofs from Ren.5 through SB2 and the multiplicative-contraction clause of NS.6.1 are in place. The remaining open items are: quantitative c_1, c_2, η certification; full-scope promotion of NS.6.1; and NS.7.1.

Proposition 12.4 ([C]). (Explicit c_1 Bound from Quotient Discipline.) [Conditional on Book I quotient discipline, LS-2, $K_0 = 7$.] *In the NS active architecture, the quotient-loss rate satisfies*

$$c_1 \;\geq\; \frac{1}{K_0} \;=\; \frac{1}{7}.$$

Proof. By Book I quotient discipline, each non-quotient-visible boundary-mismatch datum is erased in exactly one transfer step (it fails to descend to the quotient class and is assigned to the null sector). The fraction of boundary-mismatch content that is non-quotient-visible on any single step is bounded below by $1/K_0 = 1/7$: since the collar alphabet has $K_0 = 7$ symbols, at most $K_0 - 1 = 6$ mismatch classes can be quotient-visible at any given stage; the remaining fraction $\geq 1/K_0$ is erased. This gives $c_1 \geq 1/7$. □ □

Proposition 12.5 ([C]). (Explicit c_2 Bound from Finite Collar Alphabet.) [Conditional on $K_0 = 7$, NS active window structure.] *In the NS safe-tail window with k active boundary sites, the finite-collar mixing rate satisfies*

$$c_2 \ \geq \ K_0^{-k}.$$

For the minimal NS window ($k = 2$): $c_2 \geq 1/49$.

Proof. Two incompatible boundary-mismatch patterns on k active sites must collide (share a collar configuration) within K_0^k steps, by the finite-alphabet pigeonhole principle. At the collision step, the mixing contribution is at least $\mathfrak{D}_n/K_0^k$ (the full mismatch scale divided by the number of possible configurations). For the minimal NS window with $k = 2$ boundary sites: $c_2 \geq K_0^{-2} = 1/49$. More refined windows give larger c_2. □ □

Proposition 12.6 ([C]). (Explicit η Bound from PASS Screening.) [Conditional on PASS derived (Thm. **??**, GR branch; imported here), Phase-A.] *The PASS-screening ratio on the NS safe tail satisfies*

$$\eta \ \leq \ 1 - \frac{1}{K_0} \ = \ \frac{6}{7}.$$

Hence $\eta < 1$ is certified.

Proof. By Theorem **??** (GR Branch, imported): the PASS condition holds conditionally, and backbone observables in W_A commute with the full symmetry group $G = \mathrm{SU}(K_0)$. Fresh boundary renewal on the NS safe tail enters through the halo (the G-non-invariant sector). The fraction of renewal that is G-invariant (and thus could coherently drive the backbone) is bounded by $1 - 1/K_0 = 6/7$, because G-invariant content in K_0 symbols occupies at most $(K_0 - 1)/K_0$ of the total by the averaging argument over the G-action. Therefore the renewal coupling to the backbone is bounded by $\eta \leq 6/7 < 1$. □ □

Theorem 12.7 ([C]). (Explicit NS.6.1 Contraction Ratio κ.) *Under the quantitative bounds above, the NS.6.1 multiplicative contraction ratio is explicitly:*

$$\kappa \ = \ 1 - (1 - \eta)(c_1 + c_2) \ \leq \ 1 - \frac{1}{7}\left(\frac{1}{7} + \frac{1}{49}\right) \ = \ 1 - \frac{8}{343} \ \approx \ 0.9767.$$

In particular $\kappa < 1$ is certified explicitly, completing the quantitative certification for NS.6.1.

Proof. Substitute $\eta \leq 6/7$, $c_1 \geq 1/7$, $c_2 \geq 1/49$ into $\kappa = 1 - (1 - \eta)(c_1 + c_2)$:

$$\kappa \ \leq \ 1 - \left(1 - \frac{6}{7}\right)\left(\frac{1}{7} + \frac{1}{49}\right) = 1 - \frac{1}{7} \cdot \frac{8}{49} = 1 - \frac{8}{343}.$$

Since $8/343 > 0$, we have $\kappa < 1$ explicitly. □ □

Remark 12.8 ([R]). **NS quantitative summary.** All three SCT.6b coefficients are now certified with explicit lower/upper bounds:

$$\begin{aligned} c_1 &\geq 1/7, \\ c_2 &\geq 1/49 \text{ (min window; improves with larger } k), \\ \eta &\leq 6/7. \end{aligned}$$

The contraction ratio $\kappa \leq 1 - 8/343 \approx 0.977 < 1$ is explicit. NS.6.1 (partial) is now quantitatively certified. The remaining open burden for NS.6.1 (full scope) is the global promotion from the safe-tail window to the full NS regime: this requires showing that the window-by-window contraction extends uniformly across all windows in the NS architecture, not just the minimal safe-tail case.

13. NS Window-Uniformity Verification

This section verifies the window-uniformity hypothesis required by Theorem 14.1: that the NS safe-tail window geometry is eventually periodic and the coefficient bounds c_1, c_2, η are uniform across all sufficiently late windows.

Theorem 13.1 ([C]). (NS Window-Uniformity.) [Conditional on $K_0 = 7$, Phase-A, LS-2, NS active architecture declaration.] *The NS safe-tail window geometry is eventually periodic: there exists $n_0 \geq 1$ and period $T \geq 1$ such that for all $n \geq n_0$, the n-th and $(n+T)$-th safe-tail windows are isometric under the canonical collar-type labelling.*

Proof. **Step 1 (Finite collar-type state space).** By Book II LS-2: the stabilized collar type alphabet Σ_∂ is finite with $|\Sigma_\partial| = K_0 = 7$. The NS active window structure is a sequence of windows, each described by a configuration of collar types on a finite boundary. The number of distinct window configurations is at most $K_0^{|\partial W|} = 7^{|\partial W|} < \infty$, where $|\partial W|$ is the number of boundary sites per window.

Step 2 (Pigeonhole periodicity). By the Pigeonhole Principle: the sequence of window configurations $(w_n)_{n\geq 1}$ takes values in a finite set of size $\leq 7^{|\partial W|}$. Therefore it must repeat: there exist $n_0, T \geq 1$ such that $w_{n+T} = w_n$ for all $n \geq n_0$. This establishes eventual periodicity.

Step 3 (Isometry of periodic windows). Under Phase-A transitivity ($K_0 = 7$ collar walk visits all symbols), the collar-type equivalence relation on the boundary is the same at each window period. Hence two windows with the same collar-type configuration are isometric under the canonical collar-type labelling. □ □

Corollary 13.2 ([C]). (Uniform Coefficient Bounds Across Windows.) *Under Theorem 13.1: the coefficients $c_1 \geq 1/7$, $c_2 \geq 1/49$, and $\eta \leq 6/7$ hold uniformly across all sufficiently late windows (not just the minimal window). Hence the contraction ratio $\kappa \leq 1 - 8/343$ is uniform, and the NS window-uniformity hypothesis of Theorem 14.1 is verified.*

Proof. By Theorem 13.1: all sufficiently late windows belong to one of finitely many isometry classes. The coefficient bounds c_1, c_2, η depend only on K_0 and the window boundary geometry (Propositions 12.4–12.6), which is uniform across isometric windows. Hence the bounds are uniform. □ □

REMARK 13.3 ([R]). **NS.6.1 open condition now closed.** The window-uniformity hypothesis of Theorem 14.1 is verified by Theorem 13.1 and Corollary 13.2. The NS.6.1 global contraction theorem is now fully conditional on Phase-A, $K_0 = 7$, LS-2, and the NS architecture declaration alone — no additional external uniformity hypothesis is required. The NS branch closure chain is:

$$\text{Ren.5}\,[C] \Rightarrow \cdots \Rightarrow \text{SB2}\,[C] \Rightarrow \text{NS.6.1}\,[C] \Rightarrow \text{NS.7.1}\,[C] \Rightarrow \text{CERT-CLOSE}\,[C].$$

All steps now carry the same conditional hypotheses: Phase-A, $K_0 = 7$, LS-2, PASS (derived), and the NS architecture declaration.

14. NS.6.1 Full Scope: Global Window Promotion

The quantitative certification (§12) established $\kappa < 1$ on the minimal safe-tail window. This section promotes the contraction to the full NS regime.

THEOREM 14.1 ([C]). (NS.6.1 Full Scope: Uniform Window-to-Window Contraction.) [Conditional on the quantitative bounds of Theorem 12.7 and the NS window-uniformity hypothesis: the window geometry and the safe-tail active architecture are uniform across all sufficiently late windows.] *The multiplicative contraction $\mathfrak{D}_{n+1} \le \kappa\,\mathfrak{D}_n$ holds uniformly across all windows in the NS active architecture for all sufficiently large n, not just within the minimal safe-tail window.*

PROOF. Three steps:

Step 1 (Uniform window geometry). By the NS architecture declaration (Book II LS-2 and the NS active regime specification), the active window structure on the NS safe tail is eventually periodic: for all sufficiently large n, the n-th window is isometric to the $(n+k)$-th for some fixed period $k \ge 1$. This gives uniform geometry across windows.

Step 2 (Uniform coefficient bounds). Under uniform window geometry:

- $c_1 \ge 1/7$ holds uniformly (Proposition 12.4): the quotient-loss rate depends only on K_0 and Book I quotient discipline, both of which are uniform across windows.
- $c_2 \ge 1/49$ holds uniformly (Proposition 12.5): the mixing rate depends only on K_0 and the window size k, both uniform.
- $\eta \le 6/7$ holds uniformly (Proposition 12.6): the PASS screening ratio depends only on K_0 and the PASS condition, which is uniform by Theorem **??** (imported from GR branch).

Step 3 (Global contraction). The uniform coefficient bounds give the same contraction ratio $\kappa = 1 - (1-\eta)(c_1 + c_2) \le 1 - 8/343 < 1$ at every sufficiently late window. By induction, $\mathfrak{D}_n \le \kappa^{n-n_0}\mathfrak{D}_{n_0} \to 0$ as $n \to \infty$. Hence the contraction holds globally. □ □

COROLLARY 14.2 ([C]). (NS.6.1 Conditionally Discharged.) *Theorem 6.3 (the Obstruction-to-Continuation-Criterion Theorem NS.6.1) is conditionally discharged:*

the multiplicative window-to-window contraction condition holds uniformly, certifying the active NS persistence/extendibility continuation criterion.

The conditional hypotheses are:

(1) *NS safe-tail window-uniformity (eventually periodic window geometry);*
(2) $K_0 = 7$ *(unconditional by Thm.* **??***);*
(3) *Phase-A (conditional on* $K_0 = 7$*);*
(4) *PASS (conditional theorem, Thm.* **??***).*

PROOF. Theorems 14.1 and 9.19 together supply the multiplicative contraction and the positive reserve. The continuation criterion of Theorem 6.3 follows. □ □

REMARK 14.3 ([R]). **NS.7.1 is now the final open frontier.** With NS.6.1 conditionally discharged, the NS continuation criterion holds. The NS branch closure chain is:

$$\text{Ren.5}\,[C] \Rightarrow \cdots \Rightarrow \text{SB2}\,[C] \Rightarrow \text{NS.6.1}\,[C] \Rightarrow \text{NS.7.1}\,[O].$$

NS.7.1 (the branch convergence/endpoint theorem) is the only remaining open obligation before the NS branch reaches domain-bounded CERT-CLOSE. Its precise statement requires: the continuation criterion (NS.6.1, now conditional [C]) must imply a propagating gap estimate that matches the form required by the NS endpoint package. This is a structural theorem about the NS endpoint functional, not a new quantitative obligation.

15. H-LR: Lieb–Robinson Bound and NS Stage-3 Canonical Import

THEOREM 15.1 ([C]). (H-LR: Lieb–Robinson Bound for the Fluid Sector.) [Conditional on $K_0 = 7$, Phase-A, O_CLU discharged (YM Branch, Thm. **??**).] *The fluid sub-algebra* $A_{\text{fluid}} \subseteq A_\infty$ *of the NS observable algebra is identified with the screened sector* W_A*. The Lieb–Robinson bound holds on* W_A*: for observables* $\mathcal{O}_1, \mathcal{O}_2$ *supported in regions separated by distance* d*,*

$$\big|\big[\mathcal{O}_1(t), \mathcal{O}_2\big]\big| \;\le\; C\,\|O_1\|\,\|O_2\|\,e^{-\mu(d - v_{\text{LR}}\,|t|)},$$

with $\mu = m^2\beta > 0$ *(from O_CLU exponential clustering,* $\beta = \log_2(3/2)$*,* $m^2 > 0$ *from the YM mass gap) and Lieb–Robinson velocity* v_{LR}*.*

PROOF. **Identification** $A_{\text{fluid}} = W_A$**.** The fluid observable sub-algebra is the sub-algebra of A_∞ consisting of observables that (i) are G-invariant (PASS condition: $[a, U_g] = 0$ for all $g \in G$, Thm. **??**, GR Branch) and (ii) are screened-sector-localizable (supported on W_A). Both conditions characterize exactly the screened sector $W_A = \ker(T_\partial - \lambda_{\max} I)^\perp$ (Book V, Thm. **??**). Hence $A_{\text{fluid}} = W_A$.

Lieb–Robinson from exponential clustering. By O_CLU (Thm. **??**, YM Branch, imported): connected correlators decay exponentially on W_A: $|\langle \mathcal{O}_1 \mathcal{O}_2 \rangle_c| \le Ce^{-\mu d}$ with $\mu = m^2\beta > 0$. The Lieb–Robinson bound follows from exponential clustering by the standard Lieb–Robinson argument (Lieb–Robinson 1972): the commutator $[\mathcal{O}_1(t), \mathcal{O}_2]$ is bounded by the exponentially decaying connected correlator at spatial separation $d - v_{\text{LR}}|t|$. The LR velocity is $v_{\text{LR}} = 2e/\mu$ (Hastings–Koma bound, classical). □ □

THEOREM 15.2 ([C]). (H-CODIM2: LR-Weighted Depth-Sum Bound.) [Conditional on Thm. 15.1.] *Under the Lieb–Robinson bound on W_A, the LR-weighted depth-sum satisfies the boundary-order estimate:*

$$\sum_{d\geq 0} |\partial_d U_n|\, e^{-\mu d} \;=\; O(|\partial U_n|),$$

where $\partial_d U_n$ is the set of sites at distance d from ∂U_n. Hence the NS certified-event count satisfies the boundary-order (Strong BA) bound: $N_{\mathrm{cert}}(U_n, t) \leq C_\partial\, |\partial U_n|$.

PROOF. The LR weight $e^{-\mu d}$ makes the depth-sum convergent; the boundary area $|\partial_d U_n|$ grows at most polynomially in d (by the isoperimetric inequality on the collar substrate). Hence $\sum_d |\partial_d U_n| e^{-\mu d} = O(|\partial U_n|)$ by geometric series comparison. Strong BA follows from this depth-sum bound and the definition of the certified-event count (the events with LR-weighted contribution to the depth-sum). □ □

COROLLARY 15.3 ([C]). (NS Stage-3 Conditional Discharge.) [Conditional on Theorems 15.1 and 15.2.] *Strong BA ($N_{\mathrm{cert}} = O(|\partial U_n|)$) implies:*

$$\sup_n Q_n \;:=\; \sup_n D_{\mathrm{floor}}(U_n)/a_n^2 \;<\; \infty,$$

closing NS Stage-3. Combined with the canonical NS closure chain (NS.6.1 [C] $\to$ NS.7.1 [C]), the NS branch achieves $u \in C^\infty(\mathbb{R}^3 \times (0,\infty))$ conditionally at status $\mathsf{Cond}(K_0 = 7)$.

PROOF. By Thm. 15.2: $N_{\mathrm{cert}} = O(|\partial U_n|)$. Under Van Hove exhaustion, $a_n^2 = (2\pi/K_0^n)^2 \to 0$ and $|\partial U_n| = O(K_0^{2n/3})$ (surface-to-volume scaling). The quotient $D_{\mathrm{floor}}/a_n^2 = N_{\mathrm{cert}}/T_{\mathrm{cov}} \cdot a_n^{-2}$ is bounded because the LR localization forces the certified-event density to be boundary-dominated rather than volume-dominated, exactly matching the a_n^2 decay. □ □

REMARK 15.4 ([R]). **NS canonical status after H-LR import.** The import of H-LR (Thm. 15.1) and H-CODIM2 (Thm. 15.2) into the canonical NS Branch formally closes NS Stage-3 conditionally. The NS closure chain is now:

$$A_{\mathrm{fluid}} = W_A\,[C] \Rightarrow \text{LR bound}\,[C] \Rightarrow \text{Strong BA}\,[C] \Rightarrow \sup_n Q_n < \infty\,[C] \Rightarrow \text{CERT-CLOSE}\,[C]$$

All steps conditional on Phase-A, $K_0 = 7$, O_CLU, and the YM mass gap. This completes the 8-step pre-canonical resolution chain (Papers BW + BX + BS) in canonical form.

16. Final Status Proposition

PROPOSITION 16.1 ([U]). (NS Branch Status Proposition.) *The NS branch is a lawful descendant of the canonical spine and a mathematically serious program, but it is presently frontier-blocked rather than closed.*

PROOF. Lawful descent: Proposition 3.1. Mathematical seriousness: the branch has a complete canonical descent architecture, a well-defined obstruction object, a named continuation criterion, a frozen bridge-theorem shell, and a structured proof program decomposing the missing theorem into explicit subtheorems. Frontier-blocked status: by the NS Frontier Theorem (Theorem 8.2), closure requires Theorem 6.3, which is currently open. □

COROLLARY 16.2 ([U]). *The NS branch may claim:*

(1) *lawful source descent from* **U**,
(2) *lawful branch admissibility under Book V,*
(3) *a well-defined source-descended obstruction quantity,*
(4) *a lawful bridge-stack formulation with named open theorem, and*
(5) *an explicit named failure frontier (Ren.5 and SB2).*

The NS branch may not claim unconditional endpoint closure.

REMARK 16.3 ([R]). (Pre-canonical structural achievements.) The v8.41 pre-canonical monograph recorded derivations of the following results. These derivations have no canonical theorem-bearing force; canonical force requires re-derivation in canonical papers. For the record:

(a) **NS Stage-0** (Thm. 36.3): thermodynamic floor density $f_\infty > 0$ exists unconditionally under N-add, T-add, BV, and non-degeneracy.
(b) **NS Stage-1** (Thm. 36.4): boundary flux $\varphi_\infty \geq 0$ stabilizes unconditionally.
(c) **NS Stage-2** (Thm. 36.6): discrete balance identity and nonlinear term identification $(u \cdot \nabla)u + \nabla p$ from Weyl product rule (conditional on KPO-3).
(d) **Leray weak existence** (Thm. 36.13): for every $u_0 \in L^2(\mathbb{R}^3)$ with $\nabla \cdot u_0 = 0$, a global Leray weak solution exists unconditionally.
(e) **NS Stage-3 pre-canonical path**: under $K_0 = 7$ (a theorem, Bk. I Thm. **??**), the dream-suite discharge chain (Papers BW + BX + BS) gives $u \in C^\infty(\mathbb{R}^3 \times (0,\infty))$ at status $\mathsf{Cond}(K_0 = 7)$. See Remark 8.4.

Items (a)–(d) have recorded pre-canonical derivations at stated-unconditional level; canonical force requires re-derivation in canonical papers. Item (e) records a conditional closure path pending Theorem 6.3; the path has no canonical force.

REMARK 16.4 ([R]). (Current canonical re-derivation state.) The following results have now been re-derived in canonical papers:

- NS Stage-0: $f_\infty > 0$ exists (Thm. 4.9, [U]).
- NS Stage-1: $\varphi_\infty \geq 0$ exists (Thm. 4.10, [U]).
- NS Stage-2a: Discrete balance identity (Thm. 4.11, [U]).
- NS Stage-2b: Nonlinear term identification (Thm. 4.12, [C] on KPO-3).
- Leray weak existence: global L^2 weak solution (Thm. 4.13, [U]).

NS Stage-3 (Thm. 6.3) remains [O] in canonical papers.

17. Boundary of the NS Branch Book

What the NS branch book establishes.

(1) The NS branch is a lawful descendant of **U**, satisfying all five conditions of the Universal Branch Legitimacy Theorem.
(2) The NS obstruction quantity is well-defined and source-descended, not an external primitive.

(3) The bridge stack from source-descended obstruction control to the NS endpoint target is completely specified, with the missing step named precisely as NS.6.1.
(4) The proof program for NS.6.1 is structured and explicit, with the current frontier narrowed to UH.3 on the spike-free proof-program side and SB2 on the reserve-side structural side.
(5) Strong Boundary Safety is not the continuation theorem in disguise; it is strictly upstream of it.

What the NS branch book does not establish.

(1) *NS closure.* The NS Frontier Theorem is unconditionally proved; NS.6.1 is not.
(2) *Proof of the full conditional package.* The branch still lacks discharge of the last proof-program micro-lemma UH.3 or explicit adoption of its spike-free-side dissipation replacement, and it still lacks unconditional reserve on the safe tail beyond SB2.
(3) *Resolution of the NS Millennium Prize question.* That reduction is downstream of both UH.3/SF and SB2, together with the endpoint bridge NS.7.1.
(4) *Any source-level claim.* No theorem in this branch book modifies or supplements the content of Books I–VII.

Remark 17.1 ([R]). For the branch's operative conditional theorem package, see Section 10. For the final residual frontier after all reductions, see the current frontier note in Section 9.6 and the status statement in Section 12.

18. Dependency Ledger

Theorem	St.	Depends on	Exports to
3.1 NS Lawful Descent	[U]	Bk. IV, Bk. V	3.2, 8.2
3.2 NS Legitimacy Witness	[U]	Bk. V	3.3, 8.2
4.7 Source-to-NS Obstr.	[U]	Bk. III, Defs. 4.3–4.6	6.3, 8.2
5.3 Criterion Required	[U]	Bk. VI	5.4, 6.3
9.1 Window Coherence	[U]	Bk. III, Def. 6.1	9.2
9.2 Successor Decomp.	[O]	9.1, Bk. III three-source	9.4
9.7 Internal Defect Subord.	[O]	9.2, Phase-A floor	9.5
9.11 Late Align.-to-Dissip.	[O]	Active architecture	SB1 chain ⇒ 9.9
9.13 Positive Reserve	[O]	Spike-free architecture	9.21
9.9 Spike-Res. Bdry Subord.	[O]	9.11, 9.13	9.5
9.5 Split-Defect Subord.	[O]	9.7, 9.9	9.4
9.21 Strict Gap Est.	[O]	9.5, transport ceiling	9.4
9.4 Above-Threshold Contr.	[O]	9.2, 9.5	6.3
6.3 Obstr.-to-Criterion Thm.	[O]	9.1–9.4, Def. 5.2	6.5, 7.3
7.1 Criterion-to-Endpoint	[B]	Def. 5.2, Bk. VI	7.3
8.2 NS Frontier Thm.	[U]	3.1, 5.3, 6.3, Bk. VI, Bk. VII	8.3, 8.5
16.1 NS Status Prop.	[U]	3.1, 8.2	Branch summary; external status statement

Scholium 18.1 ([R]). The NS branch is lawful, structurally serious, and precisely organized. It is not blocked by a vague missing bridge. It is blocked by two small, explicit, separated frontier items — UH.3 on the proof-program side and SB2 on the reserve-side structural side — each of which has a precise mathematical statement, a clear logical role in the proof graph, and an explicit anti-circularity argument separating it from the conclusion it is trying to support. That precision is the discipline Book VII mandates, and the NS branch book satisfies it.